Cryptocurrency Volatility Prediction System Low Level Design

2.1 Data Preprocessing

- **Datetime Conversion:** pd.to_datetime() is applied to both 'date' and 'timestamp' columns to ensure proper chronological sorting.
- **Infinite Value Handling:** Replaces np.inf (often caused by dividing by zero in liquidity ratios) with NaN and removes them using .dropna().
- **Feature Scaling:** Applies StandardScaler to normalize features so that large-scale values like MarketCap do not overpower small-scale values like daily_return.

2.2 Feature Engineering

2.2.1 Volatility Target Variable

The model predicts the 7-day rolling standard deviation of daily returns: $\sigma_7 d = \sqrt{(1 / (N - 1) \sum_{i=1}^{n} (r_i - \bar{r})^2)}$

2.2.2 Liquidity and Price Features

- **Liquidity Ratio:** Calculated as Volume/ MarketCap. High ratios indicate high trading activity relative to asset size.
- **HL Spread:** High - Low/Close. Normalizes the intraday range against the price.
- **OC Change:** Close - Open/Open. Measures the "gap" or trend within a single trading day.

2.3 Model Implementation

- **Algorithm: Random Forest Regressor**. It was selected for its ability to handle non-linear relationships in financial data and its resistance to overfitting.
- **Data Splitting:** Uses a **Time-Series Split** (80% training / 20% testing) to ensure the model is evaluated on data that chronologically follows the training set.

2.4 Model Evaluation Metrics

The model's performance is tracked using:

- **MAE (Mean Absolute Error):** Average error in the volatility prediction.
- **MSE (Mean Squared Error):** Penalizes larger misses in prediction.
- **R^2 Score:** Measures how much of the volatility "movement" the model actually explains.

2.5 Deployment Design

- **Persistence:** The trained model and the fitted scaler are saved as rf_model.pkl and scaler.pkl.
- **Real-time Inference:** The app loads these files into memory, takes user input, transforms it via the scaler, and generates a prediction without needing to re-access the original training dataset.